

PROFORMA FOR BIO-DATA

1. Name and full correspondence address:

Dr. Kishore Natte
Assistant Professor
Department of Chemistry
Indian Institute of Technology Hyderabad
Kandi, 502285 Sangareddy, Telangana, India

2. Email(s) and contact number(s):

kishore.natte@chy.iith.ac.in or kishorenatte@gmail.com
0091 9121562367 (mobile)

3. Institution:

Indian Institute of Technology Hyderabad

4. Date of Birth:

18 July 1984

5. Gender: Male

6. Academic Qualification

	Degree	Year	Subject	University/Institution	% of marks
1	Ph.D.	2013	Chemistry	Technical University of Berlin	magna cum laude (excellent)

7. Ph.D thesis title, Guide's Name, Institute/Organization/University, Year of Award

"Synthesis of novel polymeric hybrid nanoparticles with enhanced interfacial and colloidal properties for biomedical studies", Prof. Joerg F. Friedrich, Technical University of Berlin, 18.01.2013

8. Work and teaching experience

S.No.	Positions held	Name of the Institute	From	To
1	Assistant Professor (13 A1 level)	Indian Institute of Technology Hyderabad	21.03.2022	present
2	Senior Scientist (E1)	CSIR-Indian Institute of Petroleum, Dehradun (Pay matrix 12)	28.08.2017	17.03.2022 (due for Principal)

				Scientist (E2) has withdrawn due to prior commitment at IITH)
3	Assistant Professor (Honorary)	Academy of Scientific and Innovative Research (AcSIR)	24-08-2018	17.03.2022
4	Postdoctoral Fellow	RWTH Aachen University, Aachen, Germany	01.01.2016	04.08.2017
5	Postdoctoral Fellow	Leibniz Institute of Catalysis (LIKAT), Rostock, Germany	01.05.2013	31.12.2015
6	Scientific Coworker	BAM-Federal Institute of Materials Research and Testing, Berlin, Germany	01.10.2012	30.04.2013
S.No.	Positions held	Name of the Institute	From	To
1	Assistant Professor (13A1)	Indian Institute of Technology, Hyderabad	21.03.2022	present
2	Senior Scientist (E1)	CSIR, Indian Institute of Petroleum, Dehradun	28.08.2017	17.03.2022
3	Postdoctoral Fellow	RWTH Aachen University, Germany	01.01.2016	04.08.2017
4	Postdoctoral Fellow	Leibniz Institute of Catalysis (LIKAT), Rostock, Germany	01.05.2013	31.12.2015
5	Scientific C-worker	BAM-Federal Institute of Materials Research and Testing	01.10.2012	30.04.2013
6	Senior Chemist	GVK Biosciences Pvt. Ltd., Hyderabad	15.05.2007	31.01.2009

9. Professional Recognition/Award/Prize/Certificate, Fellowship received by the applicant.

S.No.	Name of Award	Awarding Agency	Year
1	Marie Skłodowska Curie Fellowship (withdrawn due to prior commitment to CISR-IIP)	European Commission, Europe	2017
2	Leibniz Annex fellow	Leibniz Gemeinschaft, Germany	2013

3	Travel award to Sixteenth International Symposium On Silicon Chemistry (ISOS XVI), Hamilton, Canada	Sixteenth international symposium-on silicon chemistry-isosxvi, Canada	2011
4	BAM-Innovation Initiative Ph.D. Fellow	BAM-Federal Institute for Material Research and Testing, Berlin, Germany.	2009

12. Funding and support:

1) PI- DST SERB – “Nitrogen-doped Renewable Carbon Supported Cobalt Catalyst for Selective Hydrogenation of Arenes and Heteroarenes” (Grant No. SRG/2019/002004, 21.49 lakhs, **2019-2021**).

2) Co-PI- DST SERB - “Design and Development of Biocarbon Supported Mn and Co-based Nanocatalysts for the Transfer Hydrogenation/Deuteration Reactions using CH₃OH/CD₃OD as Hydrogen/Deuterium Source and Synthesis of Pharmaceutical Intermediates” (Grant No. CRG/2023/005734, **31,16,120 INR, 2024-2027**).

3) Co-PI MoE Stars- “Post-synthetic modification of porous organic polymers for improved CO₂ capture and its utilization for the synthesis of acrylic acid and its derivatives” (Grant No. STARS-2/2023-0065, 31 lakhs, **30-10-2023 to 29-10-2026**).

4) PI- AM Green – “Development of a selective Catalyst for Ethylene to Higher (C₈+) Hydrocarbons” (Grant No. AM GREEN/CHY/F290/2025-26/S385, 56 lakhs, **2025-2027**).

13. Mentorship:

01) Vishakha Goyal graduated – 12/12/2022

Thesis title: Heterogeneous Catalysts for Borrowing Hydrogen and Hydrogenation Reactions

02) Naina Sarki graduated – 15/03/2024

Thesis title: Transition Metal-catalysts for *N*-alkylation and Hydrogenation Reactions

(b) Ph.D. guidance: Number of completed and ongoing Ph.D. students at IITH.

01) Chitra Dewangan

(UGC-JRF), Roll no. cy22resch11024

02) Tarun Bhatt

(PMRF Fellow), Roll no. cy22resch11025

03) Sandeep Kumawat

(UGC-JRF), Roll no. cy23resch12002

04) Suravi Dey

(MoE), Roll no. cy23resch11020

05) Vadithya Suman

(UGC-JRF), Roll no. cy23resch12008

06) Sadhana Jaiswal

(CSIR-JRF), Roll no. cy24resch11003

07) K. Yaswanth

(UGC-JRF), Roll no. cy25resch01003

08) Kallakuri Leela Manikya Naga Siva Jyothi

(CSIR-JRF), Roll no. cy25resch02005

14. Publications (List of papers published in SCI Journals, in year wise descending order)

S.No.	Author(s)	Title	Name of Journal	Volume	Page	Year
76	S. Kumawat, G. N. Nair, V. N. Kalevaru, S. Wohlrab, T. Tarasiuk, T. Yegorova, P. K. Mykhailiuk, K. Natte*	N-Selective Difluoromethylation of 4-Hydroxyquinolines	Organic Letters	27 (41)	11466–11473	2025
75	S. Kumawat, K. L. M. N. S. Jyothi, V. N. Kalevaru, S. Wohlrab, K. Natte*	DAST Enabled Synthesis of Fluorinated Amides and Fatty Acid Amides Including Drugs under Ambient Conditions	Organic Letters	27(32)	8829-8834	2025
74	V. Suman, T. Bhatt, G. N. Nair, V. N. Kalevaru, S. Wohlrab, K. Natte*	CO ₂ Valorization toward Selective N-Formylation of Nitroarenes and Aminoarenes under Homogeneous Titanium Catalysis	ACS Sustainable Chemistry & Engineering	13, 40	16971–1698	2025
73	S. Kumawat, T. Bhatt, K. Natte*	Dearomative Difluoromethylation of N-heterocycles and	J. Org. Chem.	90 (28),	9826-9844	2025

		Pharmaceuticals with Bromo(difluoro)acetic acid				
72	S. Kumawat, T. Bhatt, V. Goyal, H. Neumann, R. Zbořil, R. V Jagadeesh, K. Natte*	Advances in trifluoromethylation of C (sp ³)-H bonds	Coordination Chemistry Reviews	542	216885	2025
71	C. Dewangan, S. Sarmah, K. Natte*	Nickel-Catalyzed Synthesis of Structurally Diverse N-(hetero) aryl Pyrroles	Asian J. Org. Chem.		e00195	2025 doi.org/10.1002/a-joc.202500195
70	K. Sadhana, S. Kumawat, K. L. M. N. S. Jyothi, K. Natte*	Silver (I) Oxide Catalyzed C/S–H Trifluoromethylation of Arenes and Heteroarenes with Sodium Trifluoromethanesulfinate	Synlett			2025 DOI: 10.1055/s-0043-1775481
69	T. Bhatt, V. Suman, M. Choudhary, S. K. Singh, K. Natte*	Dearomative selective reduction of structurally diverse N-heteroarenes enabled by a homogeneous titanium catalyst	Journal of Catalysis	443	115937	2025
68	V. Goyal, T. Bhatt, A. Kuttasseri, A. Mahata, R. Zbořil, K. Natte , R.V. Jagadeesh	A biomass-derived nickel-based nanomaterial as a sustainable and reusable catalyst for hydrogenation of arenes and heteroarenes	RSC Sustainability	3	2235	2025
67	V. Vakati, V. Goyal, A. S Alshammari, N. V Kalevaru, S. Wohlrab, K. Natte , R.	Practical and scalable hydrogenation of nitro compounds using palladium-based nanocatalyst under ambient conditions	Catalysis Today	442	114940	2024

	Zbořil, R. V. Jagadeesh					
66	S. Kumawat, <u>K. Natte*</u>	Accessing Structurally Diverse Aryl Difluoromethyl Ethers with Bromo(difluoro)acetic Acid	Chem Commun			2024 (just accepted) DOI doi.org/10 .1039/D4 CC0478 2F
65	S. Kumawat, S. Dey, <u>K. Natte*</u>	Ammonia-Borane Dependent Transfer Hydrogenation of Carboxylic Esters to Primary Alcohols	J. Org. Chem	89	10719– 10728	2024
64	S. Kumawat, S. Singh, T. Bhatt, T. Maurya, S. Vaidyanathan , <u>K. Natte,*</u> J. Rajenahally	Valorization of Bio- Renewable Glycerol by Catalytic Amination Reactions	Green Chem	26	3021- 3038	2024
63	S. Kumawat, <u>K. Natte*</u>	Iron (II) triflate as a photocatalyst for trifluoromethylation of functionalized arenes under blue LED light: Access to bioactive compounds	J. Catal.	434	115506	2024
62	T. Bhatt, <u>K. Natte*</u>	Transfer Hydrogenation of N- and O-containing heterocycles Including Pyridines with H ₃ N– BH ₃ Under the Catalysis of the Homogeneous Ruthenium Pre-catalyst	Org. Lett.	26	866–871	2024
61	N. Sarki, S. Kumawat, M. Choudhary, A. Narani, S. K. Singh, <u>K. Natte*</u>	Selective hydrogenation of carbonyl compounds and olefins using an efficient combination of homogeneous	J. Catal.	429	115248	2023

		RuCl ₃ /1,10-Phenanthroline				
60	C. Dewangan, S. Kumawat, T. Bhatt, K. Natte*	Homogenous Nickel-Catalyzed Chemoselective Transfer Hydrogenation of Functionalized Nitroarenes with Ammonia-Borane	Chem Commun.	59	14709-14712	2023
59	B. Singh, V. Goyal, D. Sarma, R. Kumar, T. Bhatt, A. Mahata, C. R. Reddy, A. Narani, K. Natte*	Earth-Abundant Heterogeneous Cobalt Catalyst for Selective Ring Hydrogenation of (Hetero) arenes and Gram-Scale Synthesis of Pharmaceutical Intermediates	ACS Catal.	13	9724-9744	2023
58	M. Subramanian, G. Sivakumar, V. G Landge, R. Kumar, K. Natte , R. V Jagadeesh, E. Balaraman	General and selective homogeneous Ru-catalyzed transfer hydrogenation, deuteration, and methylation of functional compounds using methanol	J. Catal.	425	386-405	2023
57	V. Goyal, T. Bhatt, C. Dewangan, A. Narani, G. Naik, E. Balaraman, K. Natte ,* R. V. Jagadeesh,	Methanol as a Potential Hydrogen Source for Reduction Reactions Enabled by a Commercial Pt/C Catalyst	J. Org. Chem	88	2245–2259	2023
56	R. Cauwenbergh, V. Goyal, R. Maiti, K. Natte ,* S. Das	Challenges and Recent Advancements in the Transformation of CO ₂ to Carboxylic acids: Straightforward Assembly with homogeneous 3d Metals	Chem. Soc. Rev	51	9371-9423	2022

54	V. Goyal, N. Sarki, A. Narani, G. Naik, K. Natte* , R. V. Jagadeesh*	Recent Advances in the Catalytic N-Methylation and N-Trideuteromethylation Reactions Using Methanol and Deuterated Methanol	Coordination Chemistry Reviews	474	214827	2023
53	T. Senthamarai, F. Poovan, A. M. Alenad, N. Rockstroh, J. Rabeah, S. Bartling, E. Baráth, K. Natte , R. V. Jagadeesh,	Cu-Oxide Nanoparticles Catalyzed Synthesis of Nitriles and Amides from Alcohols and Ammonia in Presence of Air	Adv. Sustainable. Syst., 2022,	6	2200263	2022
52	B. Singh, R. Kumar, N. Singh, D. Tripathi, K. Natte* , A. Narani	Hydrogenation of Lignin-Derived Feedstocks and Bio-oil using Active and Stable Ruthenium Catalyst,	Catalysis Today	408	139-149	2023
51	N. Sarki, A. Narani, G. Naik, D. Tripathi, S. L. Jain, K. Natte*	Biowaste Carbon Supported Manganese Nanoparticles as an Active Catalyst for the Selective Hydrogenation of Bio-based Aldehydes	Catalysis Today	408	127-138	2023
50	N. Sarki, R. Kumar, B. Singh, A. Ray, G. Naik, K. Natte* , A. Narani	Lignin Residue-Derived Carbon-Supported Nanoscale Iron Catalyst for the Selective Hydrogenation of Nitroarenes and Aromatic Aldehydes	ACS Omega	23	19804–19815	2022
49	F. Poovan, V. Chandrashekar, K. Natte* , J. Rajenahally	Synergy between homogeneous and heterogeneous catalysis.	Catal. Sci. Technol	12	6623-6649	2022
48	G. Naik, N. Sarki, V. Goyal, A.	Recent Trends in Upgrading of CO ₂ as a C1 Reactant in N- and	Asian J. Org. Chem.	11	e202200270	2022

	Narani, K. Natte,*	C-Methylation Reactions				
47	S. Rawat, B. Singh, B. Raju, C. Pendem, S. Bhandari, K. Natte,* A. Narani	Value addition of Lignin to Zingerone using recyclable AlPO ₄ and Ni/LRC Catalysts,	Chem. Eng. J	431	134130	2022
46	V. G. Chandrashekar, K. Natte , A. M. Alenad, A. S. Alshammari, R. V. Jagadeesh	Reductive amination, hydrogenation and hydrodeoxygenation of 5-hydroxymethylfurfural using silica-supported cobalt-nanoparticles	ChemCatChem	14	e202101234	2021
45	N. Sarki, V. Goyal, K. Natte,* R. V. Jagadeesh	Base metal-catalyzed C-methylation reactions using methanol	Adv. Synth. Catal.	363	5028-5046	2021
44	V. Goyal, G. Naik, A. Narani, K. Natte,* R. V. Jagadeesh	Recent developments in reductive N-methylation with base-metal catalysts	Tetrahedron	98	132414	2021
43	V. Goyal, N. Sarki, M. K. Poddar, A. Narani, D. Tripathi, A. Ray, K. Natte,*	Bio renewable carbon-supported Ru catalyst for N-Alkylation of Amines with Alcohols and Selective Hydrogenation of Nitroarenes	New J Chem	45	14687-14694	2021
42	V. Goyal, N. Sarki, K. Natte , A. Ray,	Pd/C-catalyzed transfer hydrogenation of aromatic nitro compounds using methanol as a hydrogen source	J. Indian Chem. Soc	98	100014	2021
41	N. Sarki, V. Goyal, N. K. Tyagi, Puttaswamy, A. Narani, A. Ray, K. Natte*	Simple RuCl ₃ -catalyzed N-Methylation of Amines and Transfer Hydrogenation of Nitroarenes using Methanol	ChemCatChem	7	1722-1729	2021

40	V. Goyal, N. Sarki, B. Singh, A. Ray, M. Poddar, A. Bordoloi, A. Narani, K. Natte*	Carbon-Supported Cobalt Nanoparticles as Catalysts for the Selective Hydrogenation of Nitroarenes to Arylamines and Pharmaceuticals	ACS Appl. Nano Mater.	11	11070–11079	2020
39	A. Kumar, V. Goyal, N. Sarki, B. Singh, A. Ray, T. Bhaskar, A. Bordoloi, A. Narani, K. Natte*	Biocarbon supported ruthenium-oxide based catalyst for hydrogenation of arenes and heteroarenes	ACS Sustainable Chem. Eng.	41	15740–15754	2020
38	K. Natte* , A. Narani, V. Goyal, N. Sarki, R. V Jagadeesh	Synthesis of Functional Chemicals from Lignin-derived Monomers by Selective Organic Transformations	Adv. Synth. Catal.	23	5143–5169	2020
37	A. Narani, H.P.R. Kannapu, K. Natte , D. R Burri	Pd-Nanoparticles immobilized organo-functionalized SBA-15: An efficient heterogeneous catalyst for selective hydrogenation of C-C double bonds of α , β -unsaturated carbonyl compounds	Mol. Catal	497	111200	2020
36	S. Rawat, P. Gupta, B. Singh, T. Bhaskar, K. Natte* , A. Narani*	Molybdenum-catalyzed oxidative depolymerization of alkali lignin: Selective production of Vanillin,	Appl. Catal. A-Gen.	598	117567	2020
35	K. Murugesan, T. Senthamarai, V. Chandrashekar, K.	Catalytic reductive aminations using molecular hydrogen for synthesis of different kinds of amines	Chem. Soc. Rev.	49	6273–6328	2020

	Natte, P. Kamer, M. Beller, J. Rajenahally					
34	A. S Alshammari, K. Natte , N. V. Kalevaru, A. Bagabas, R. V Jagadeesh	Scalable preparation of stable and reusable silica-supported palladium nanoparticles as catalysts for N- alkylation of amines with alcohols	J. Catal.	382	141-149	2020
33	V. Goyal, J. Gahtori, A. Narani, P. Gupta, A. Bordoloi, K. Natte*	Commercial Pd/C- Catalyzed N- Methylation of Nitroarenes and Amines Using Methanol as Both C1 and H2 Source	J. Org. Chem., 23	23	15389– 15398	2019
32	T. Senthamarai, K. Murugesan, K. Natte , N. Kalevaru, H. Neumann, P. Kamer, J. Rajenahally	Expedient Synthesis of N-Methyl- and N- Alkylamines by Reductive Amination using Reusable Cobalt Oxide Nanoparticles	ChemCatChem (highlighted as a Front Cover)	10	1235- 1240	2018
31	K. Natte , R. V. Jagadeesh, H. Neumann, M. Beller	Convenient Reductive Aminations without Hydrogen: Selective Iron-catalyzed Synthesis of N- Methylamines	Nature Communicatio ns	8	1344- 1353	2017
30	K. Natte , R. K. Natte, R. V. Jagadeesh, H. Neumann, M. Beller	Transition Metal- Catalyzed Utilization of Methanol as a C1 Source in Organic Synthesis	Angew. Chem. Int. Ed	56	6384- 6394	2017
29	K. Natte , R. V. Jagadeesh, M. Sharif, H. Neumann and M. Beller	Synthesis of Nitriles from Amines using Nanoscale Co ₃ O ₄ - based Catalysts via Sustainable Aerobic Oxidation	Org. Biomol. Chem	14	3356- 3359	2016

28	<u>K. Natte</u> , R. V. Jagadeesh, L. He, J. Rabeah, C. Taeschler, H. Neumann, A. Brückner and M. Beller	Palladium Catalyzed Trifluoromethylation of Hetero(arenes) with CF ₃ Br	Angew. Chem. Int. Ed	55	2782-2786	2016
27	J. Chen, <u>K. Natte</u> , X-F. Wu	Palladium-catalyzed carbonylative C–H activation of arenes with norbornene as the coupling partner	Journal of Organometallic Chemistry	803	9-12	2016
26	X-F. Wu, <u>K. Natte</u>	The Applications of DMSO as Reagent in Organic Synthesis	Adv. Synth. Catal	358	336-352	2016
25	L. He, <u>K. Natte</u> , J. Rabeah, C. Taeschler, H. Neumann, A. Brückner and M. Beller,	Heterogeneous Pt Catalyzed C-H Perfluoroalkylation of Arenes and Heteroarenes	Angew. Chem. Int. Ed	54	4320-4324	2015
24	R. V. Jagadeesh, <u>K. Natte</u> , H. Junge, and M. Beller	Nitrogen-Doped Graphene Activated Iron Oxide-based Nanocatalysts for Selective Transfer Hydrogenation of Nitroarenes	ACS Catal.	5	1526-1529	2015
23	<u>K. Natte</u> , H. Neumann, X-F. Wu	Pd/C as an Efficient Heterogeneous Catalyst for Carbonylative Four Component Synthesis of 4(3H)-Quinazolinones	Catal. Sci. Technol	5	4474–4480	2015
22	J. Chen, J. Feng, <u>K. Natte</u> , X.-F. Wu	Palladium-Catalyzed Carbonylative Cyclization of Arenes via C-H	Chem. Eur. J	21	16370–16373	2015

		Bond Activation with DMF as the Carbonyl Source				
21	J. Chen, <u>K. Natte</u> , X-F. Wu	Pd/C-catalyzed carbonylative C–H activation with DMF as the CO source	Tetrahedron Letters	56	6413–6416	2015
20	<u>K. Natte</u> , W. Li, S. Zhou, H. Neumann, X-F. Wu	Iron-catalyzed reduction of aromatic aldehydes with paraformaldehyde and H ₂ O as the hydrogen source	Tetrahedron Letters	56	1118–11121	2015
19	J. Chen, <u>K. Natte</u> , N. Y. T Man, S. G Stewart, X-F. Wu	Convenient copper-mediated Chan–Lam coupling of 2-aminopyridine: facile synthesis of N-arylpyridin-2-amines	Tetrahedron Letters	56	4843–4847	2015
18	J. Chen, <u>K. Natte</u> , H. Neumann, and Xiao-Feng Wu	Convenient Palladium-Catalyzed Carbonylative Synthesis of Caprolactam and Butyrolactam Derived Phthalimides and Amides by using DBU and DBN as the Nitrogen Source	Tetrahedron Letters	56	342–345	2015
17	<u>K. Natte</u> , A. Dumrath, H. Neumann, M. Beller	Palladium-Catalyzed Carbonylations of Aryl Bromides by using Paraformaldehyde: Synthesis of aldehydes and esters	Angew. Chem. Int. Ed	53	10090–10094.	2014
16	J. Chen, <u>K. Natte</u> , A. Spannenberg, H. Neumann, P. Langer, M.	Base-Controlled Selectivity toward Linear and Angular Fused Quinazolinone	Angew. Chem. Int. Ed	53	7579–7583	2014

	Beller and Xiao-Feng Wu	Cores via Palladium-Catalyzed Carbonylation/Nucleophilic Aromatic Substitution Sequence				
15	K. Natte , J. Chen, H. Neumann, M. Beller and Xiao-Feng Wu	Palladium-Catalyzed Oxidative Carbonylative Coupling of Arylboronic acids with Terminal Alkynes to Alkynones	Org. Biomol. Chem	12	5590–5593	2014
14	K. Natte , J. Chen, H. Neumann, M. Beller and Xiao-Feng Wu,	Palladium-Catalyzed Carbonylation of 2-Bromoanilines with 2-Formylbenzoic acid and 2-halobenzaldehydes: Efficient Synthesis of Functionalized Isoindolinones	Chem. Eur. J	20	14184–14188	2014
13	J. Chen, K. Natte , A. Spannenberg, H. Neumann, M. Beller and Xiao-Feng Wu	Efficient Palladium Catalyzed Double Carbonylation of o-Dibromobenzenes: Synthesis of Thalidomide	Org. Biomol. Chem. (highlighted as cover picture)	12	5578-5581	2014
12	J. Chen, K. Natte , H. Neumann, M. Beller and Xiao-Feng Wu	Palladium-Catalyzed Carbonylative [3+2+1] Annulation of N-Arylpyridine-2-amines with Internal Alkynes via C-H Activation: Facile Synthesis of 2-Quinolinones	Chem. Eur. J	20	14189–14193	2014
11	J. Chen, L. He, K. Natte , H. Neumann, M. Beller and Xiao-Feng Wu	Palladium@Cerium(IV) oxide Catalyzed Oxidative Synthesis of N-(2-Pyridyl)Indoles via C-H Activation Reaction	Adv. Synth. Catal	356	2955–2959	2014

10	J. Chen, <u>K. Natte</u> , H. Neumann, and Xiao-Feng Wu	Palladium-Catalyzed Carbonylative Reactions of 1-Bromo-2-fluorobenzenes with Various Nucleophiles: Effective Combination of Carbonylation and Nucleophilic Substitution	Chem. Eur. J	20	16107–16110	2014
09	J. Chen, <u>K. Natte</u> , H. Neumann, M. Beller and Xiao-Feng Wu	A Convenient Palladium-Catalyzed Carbonylative Synthesis of Quinazolines from 2-Aminobenzylamine and Aryl Bromides	RSC Adv	4	56502–56505	2014
08	S. Rades V-D. Hodoroaba, T. Wirth, M. P. Lobera, R. H. Labrador, <u>K. Natte</u> , T. Behnke, T. Grossa, W. E. S. Unger	Morphological and elemental analysis of single engineered nanoparticles by high-resolution imaging with SEM/T-SEM, EDX and SAM	RSC Adv	4	49577–49587	2014
07	<u>K. Natte</u> , J. Friedrich, S. Wohlrab, J. Lutzki, R. von Klitzing, W. Österle and G. Orts-Gil,	Impact of polymer shell on the formation and time evolution of nanoparticle-protein corona	Colloids and Surfaces B: Biointerfaces	104	213–220	2013
06	G. Orts-Gil, <u>K. Natte</u> , R. Thiermann, M. Girod, A. Thünemann, M. Maskos and W. Österle	On the role of protein corona formation and evolution in the colloidal stability of nanoparticles	Colloids and Surfaces B: Biointerfaces	108	110–119	2013

05	G. Orts-Gil, K. Natte , W. Österle	Multi-parametric reference nanomaterials for toxicology: state of the art, future challenges and potential candidates	RSC Adv	3	18202–18215	2013
04	V. Hodoroba, S. Rades, K. Natte , G. Orts-Gil, W. S. Unger.	Morphology of Nanoparticles- A Characterization using High-Resolution SEM	Imaging and Microscopy	13	54–56	2013
03	K. Natte , G. Orts-Gil, J. Friedrich, R. von Klitzing, W. Österle	Tuning interfacial properties and colloidal behavior of hybrid nanoparticles by controlling the polymer precursor	Macromol. Chem. Phys.	212	2412–2419	2012
02	K. Natte , T. Bhenke, G. Orts-Gil, C. Würth, U. R. Genger, J. Friedrich, W. Österle	Synthesis and characterization of highly fluorescent core-shell nanoparticles based on Alexa-Dyes	J. Nanopart. Res	14	680-689	2012
01	G. Orts-Gil, K. Natte , D. Drescher, H. Bresch, A. Mantion, J. Kneipp, W. Österle	Characterization of silica nanoparticles prior to in vitro studies: from primary particles to agglomerate	J. Nanopart. Res	13	1593–1604.	2011

15. Detail of patents

S.No.	Patent Title	Name of Applicant(s)	Patent No.	Award Date	Agency/ Country	Status
1	DIFLUOROMETHYLATION OF COMPLEX N-HETEROCYCLES	K. Natte and S. Kumawat	Indian Patent Application No. 202441071161			filed
2	A method for chemoselective	K. Natte and T. Bhatt	565327	28.09.2024	India	Granted

	reduction of N-heteroarenes and its hydrogenated products					
3	DIFLUOROMETHYLATION OF COMPLEX N-HETEROCYCLES	K. Natte and S. Kumawat	569964	22-08-2025	India	Granted
2	Method for preparation of alkylated or fluoro, chloro and fluoroalkylated compounds by heterogeneous catalysis	C. Taeschler, F. Zaragoza, M. Beller, H. Neumann, L He, <u>K. Natte</u>	9771367 B2	26.09.2017	United States Patent	Granted
3	Method for preparation of alkylated or fluoro, chloro and fluoroalkylated compounds by homogeneous catalysis	C. Taeschler, M. Beller, H. Neumann, L He, <u>K. Natte</u> , F. Zaragoza, S. Ellinger	9809596 B1	07.11.2017	United States Patent	Granted

16. Books/Reports/Chapters/General articles etc.

S.No.	Title	Author's Name	Publisher	Year of Publication
1	NHC-catalyzed CO ₂ Fixations in Organic Synthesis” from the book CO ₂ as Building Block in Organic Synthesis	V. Goyal, N. Sarki, A. Narani, <u>K. Natte*</u>	Wiley VCH Verlag GmbH	2020
2	Amines by Reduction	<u>K. Natte</u> , R. V. Jagadeesh	Kirk-Othmer Encyclopedia of Chemical Technology, 2021 John Wiley & Sons	2021
3	Chapter 21 - Amination of	N Sarki, V Goyal, A Narani, RV	Elsevier	2022

	biomass to nitrogen-containing compounds	Jagadeesh, <u>K Natte</u>		
4	Chapter 18 - Thermochemical methods for upgrading of lignin to aromatic chemicals	B Singh, S Rawat, <u>K Natte</u> , A Narani	Elsevier	2022
5	Chapter 10 Surface-modified nanomaterials for the synthesis of pharmaceuticals	<u>K. Natte</u> , R. V Jagadeesh	Elsevier	2022 251-266

References

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